

Rheumatoid arthritis

Josef S Smolen, Daniel Aletaha, Iain B McInnes



Rheumatoid arthritis is a chronic inflammatory joint disease, which can cause cartilage and bone damage as well as disability. Early diagnosis is key to optimal therapeutic success, particularly in patients with well-characterised risk factors for poor outcomes such as high disease activity, presence of autoantibodies, and early joint damage. Treatment algorithms involve measuring disease activity with composite indices, applying a treatment-to-target strategy, and use of conventional, biological, and new non-biological disease-modifying antirheumatic drugs. After the treatment target of stringent remission (or at least low disease activity) is maintained, dose reduction should be attempted. Although the prospects for most patients are now favourable, many still do not respond to current therapies. Accordingly, new therapies are urgently required. In this Seminar, we describe current insights into genetics and aetiology, pathophysiology, epidemiology, assessment, therapeutic agents, and treatment strategies together with unmet needs of patients with rheumatoid arthritis.

Introduction

Rheumatoid arthritis is one of the most prevalent chronic inflammatory diseases. It primarily involves the joints, but should be considered a syndrome that includes extra-articular manifestations, such as rheumatoid nodules, pulmonary involvement or vasculitis, and systemic comorbidities. A therapeutic revolution in the treatment of rheumatoid arthritis in the past decade—with the advent of novel therapeutics, introduction of early therapy, development of new classification criteria, and application of new effective treatment strategies—has transformed articular and systemic outcomes.^{1–6} In this Seminar, we highlight recent insights into most aspects of rheumatoid arthritis, from diagnosis to treatment strategies, and from aetiology to novel therapies. There is still a considerable unmet need in rheumatoid arthritis; full or stringent remission is not typical, nor is it usually sustained without continuing treatment, and as such it should now be the priority of research efforts.

Epidemiology, genetics, and aetiology

Rheumatoid arthritis is a chronic disease that carries a substantial burden for both the individual and society.⁷ The individual burden results from musculoskeletal deficits, with attendant decline in physical function, quality of life, and cumulative comorbid risk.⁸ The socioeconomic burden, aside from major direct medical costs, is a consequence of functional disability, reduced work capacity, and decreased societal participation.⁹ Efforts to establish the diagnosis early, initiate treatment promptly, and design novel treatment strategies to control inflammation and reduce or prevent consequent damage are paramount.

Rheumatoid arthritis has an incidence of 0.5% to 1%, with an apparent reduction from north to south (in the northern hemisphere) and from urban to rural areas.^{10,11} Some Native American populations have a very high prevalence.¹⁰ A positive family history increases the risk of rheumatoid arthritis roughly three to five times; concordance rates in twins are increased, implicating genetic factors in pathogenesis.^{10,12} The heritability of rheumatoid arthritis is currently estimated as 40–65% for seropositive

rheumatoid arthritis, but lower (20%) for seronegative disease.^{13,14}

Modern genetic technologies combined with large, well-characterised clinical cohorts have advanced our understanding of the genetics of the disease. Genome-wide association studies using single nucleotide polymorphisms have characterised more than a hundred loci associated with rheumatoid arthritis risk, most of which implicate immune mechanisms (figure 1), some of which are shared with other chronic inflammatory diseases.¹⁵ The HLA system (particularly HLA-DRB1) remains the dominant influence, strongly implicating peptide (and self-peptide) binding in pathogenesis.¹⁶ Disease-associated alleles share common amino acid sequences in the peptide-binding groove (the so-called shared epitope).¹⁷ Moreover, some HLA genotypes particularly associate with more aggressive erosive disease and with higher mortality, pointing to a crucial role of peptide binding.¹⁸

Other genetic loci probably contribute smaller functional effects that are presumably singly or cumulatively mediated,¹⁹ for example, via altered costimulatory pathways (eg, CD28, CD40), cytokine signalling, lymphocyte receptor activation threshold (eg, PTPN22), and innate immune activation (figure 1). The increased risk for rheumatoid arthritis in patients with the shared epitope is linked with seropositivity for autoantibodies against citrullinated peptides (ACPAs) and autoantibodies against IgG (rheumatoid factor [RF]). These characteristic autoantibodies for rheumatoid

Lancet 2016; 388: 2023–38

Published Online

May 3, 2016

[http://dx.doi.org/10.1016/S0140-6736\(16\)30173-8](http://dx.doi.org/10.1016/S0140-6736(16)30173-8)

This online publication has been corrected. The corrected version first appeared at thelancet.com on June 10, 2016

Division of Rheumatology, Department of Medicine 3, Medical University of Vienna, Vienna, Austria (Prof J S Smolen MD, D Aletaha MD); 2nd Department of Medicine, Hietzing Hospital Vienna, Vienna, Austria (Prof J S Smolen); and Institute of Infection, Immunity and Inflammation, University of Glasgow, Glasgow, UK (Prof I B McInnes MD)

Correspondence to:

Prof Josef S Smolen, Division of Rheumatology, Department of Medicine 3, Medical University of Vienna, A-1090 Vienna, Austria josef.smolen@wienkav.at

Search strategy and selection criteria

We searched MEDLINE using the terms “rheumatoid arthritis” in conjunction with “diagnosis”, “classification”, “epidemiology”, and “pathogenesis”. For treatment, we used recent systematic literature searches, and updated the respective searches in October, 2015, including terms on novel therapies and “treatment strategy”. Selection of articles was based on our personal judgment of relevance within the scope of this Seminar.